

EDA & Initial Results

Side-Aware Bayesian Wallet Skill Decomposition

MSc Computer Science – Thesis

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Abstract

This report presents the exploratory data analysis (EDA) of the `perpetuals_knowledge_graph` database, comprising 40.5 million perpetual futures log events across 447 599 wallets over 491 days (2025-02-19 to 2026-06-26). We characterise the wallet population, assess the feasibility of Bayesian hierarchical modelling given the trade-count distribution, test the core orthogonality hypothesis (Long vs. Short skill), and evaluate leverage, sizing, and cross-asset effects. The results confirm the central research premise—Long and Short skill show only moderate correlation ($r = 0.336$), motivating side-aware decomposition—while revealing that the existing baseline composite score is severely compressed (0.969–0.988 out of 1.0) and partially redundant. We identify 4 500 wallets with ≥ 20 closed trades as the Bayesian target population. We then implement the empirical-Bayes hierarchical model (five skill dimensions) and evaluate it against eight baselines. A key methodological finding is that the standard ≥ 5 -future-trade evaluation label is $\approx 99\%$ binomial sampling noise (reliability $\rho = 0.013$), which fully explains the apparent “luck dominates” null result; on reliably-labelled wallets the signal is strong. Under rolling multi-fold walk-forward, every Bayesian posterior score beats raw-ROI ranking by $2\text{--}3\times$ ($r_s \approx 0.34\text{--}0.48$ vs. 0.17), with timing (hit-rate) skill strongest and a learned LambdaMART ranker giving the most regime-stable ranking. The single-fold “ROI wins” finding does not survive proper walk-forward.

1 Problem Formulation

1.1 Why a Single Composite Score Fails

Every incumbent wallet-scoring system — and the baseline in this database (Script 03, `daily_trader_rankings`) — reduces a wallet w ’s entire trading history to one scalar:

$$S_w = 0.25 \cdot \text{RR}_w + 0.25 \cdot \text{WLH}_w + 0.25 \cdot \text{WLR}_w + 0.25 \cdot \text{WP}_w,$$

an equally-weighted sum of risk/reward ratio, win/loss holding-time ratio, win/loss ROI ratio, and win percentage. Section 8 shows this is empirically pathological: two of the four components are correlated at $r = 0.652$ (redundant variance counted twice) and the resulting score is compressed into a 0.969–0.988 band, making rank order nearly meaningless. The deeper issue is conceptual, not just numerical: a scalar S_w cannot distinguish *why* a wallet performs well — good entries, good exits, correct directional calls, disciplined position sizing, or simply an un-copied idea — which is exactly the information a copy-trader needs.

1.2 Formal Decomposition

Let wallet w ’s closed-position history be the ordered sequence $\{(a_i, d_i, \ell_i, x_i, y_i)\}_{i=1}^{N_w}$, where a_i is the traded asset, $d_i \in \{+1, -1\}$ the side (Long/Short), ℓ_i the leverage, and x_i, y_i the entry/exit price of trade i . The realized return of trade i is

$$r_i = d_i \frac{y_i - x_i}{x_i}.$$

Timing (hit) skill is the fraction of directionally correct trades:

$$\text{HR}_w = \frac{1}{N_w} \sum_{i=1}^{N_w} \mathbf{1}[r_i > 0].$$

Sizing skill is the win/loss ratio — whether size is larger on correct calls:

$$\text{WL}_w = \frac{\mathbb{E}[r_i \mid r_i > 0]}{|\mathbb{E}[r_i \mid r_i \leq 0]|}.$$

These two are not independent contributions to risk-adjusted return; they multiply. **Van Loon (2018)** “*Timing versus Sizing Skill*,” *Journal of Portfolio Management*, doi:10.3905/jpm.2018.44.3.025 — the original paper this decomposition is taken from — shows that the Information Ratio factors exactly as

$$\text{IR}_w = \text{HR}_w \times \text{WL}_w \times \sqrt{\text{Breadth}_w},$$

where Breadth_w is the number of independent bets. Van Loon shows empirically that timing contributes roughly twice the explanatory power of sizing, and — critically for a wallet with a $< 50\%$ hit rate — a positive IR is still achievable if WL_w is large enough. Equation 1.2 is the mathematical justification for treating timing and sizing as *separate, multiplicatively-combined* dimensions rather than folding both into one ratio, which is what the current baseline’s WLR and WP components implicitly (and redundantly) attempt.

Orthogonal to timing/sizing, **Lim et al. (2022)**, doi:10.1007/s11156-022-01065-9, show for mutual funds that entry (buy) skill and exit (sell) skill are only weakly correlated and that sell skill dominates aggregate alpha. Applied here, define per-side skill as the mean alpha over a same-asset, same-period benchmark $\bar{r}_{a,t}$:

$$s_w^{\text{Long}} = \mathbb{E}[r_i - \bar{r}_{a,t_i} \mid d_i = +1], \quad s_w^{\text{Short}} = \mathbb{E}[r_i - \bar{r}_{a,t_i} \mid d_i = -1].$$

Section 5 tests exactly this: whether s_w^{Long} and s_w^{Short} (operationalized as long/short win rate) are empirically orthogonal, i.e. $|\text{corr}(s^{\text{Long}}, s^{\text{Short}})| < 0.3$.

1.3 The Estimation Problem

Point estimates $\widehat{\text{HR}}_w, \widehat{\text{WL}}_w, \hat{s}_w^{\text{Long}}, \hat{s}_w^{\text{Short}}$ are unreliable when N_w is small — the median wallet in this database has only 3 closed trades (Section 3.3). We therefore treat each skill dimension $d \in \{\text{buy, sell, timing, sizing}\}$ as a latent variable under a hierarchical prior, following Kosowski et al. (2006) and Berk & van Binsbergen (2015):

$$\theta_w^{(d)} \sim \mathcal{N}(\mu^{(d)}, \tau^{(d)2}), \quad \hat{s}_w^{(d)} \mid \theta_w^{(d)} \sim \mathcal{N}\left(\theta_w^{(d)}, \frac{\sigma^{(d)2}}{N_w}\right),$$

with $\mu^{(d)}, \tau^{(d)}$ estimated from the wallet population. The posterior $p(\theta_w^{(d)} \mid \hat{s}_w^{(d)}, N_w)$ shrinks low- N_w wallets toward the population mean and yields a calibrated credible interval, replacing the false precision of a single scalar S_w with an uncertainty-aware, multi-dimensional skill profile.

Research question this formulation exists to answer: does replacing S_w with $\{\theta_w^{(d)}\}_d$ produce better out-of-sample ranking of copy-trading targets than the current baseline (B1) or naive PnL/ROI/win-rate rankings (B2–B4)? Section 9.3 gives the first evidence: even simple point-estimate versions of these features (overall_win_rate, long_win_rate, etc.) already show a 75% out-of-sample signal decay relative to in-sample correlation, which is precisely the small- N_w shrinkage problem this Bayesian formulation is designed to correct.

2 Data Overview

The primary data source is the `logs` collection of the `perpetuals_knowledge_graph` MongoDB, which records every lifecycle event for perpetual futures positions across five platforms and five chains.

Table 1: Data scale across collections.

Metric	Value	Source (Script)
Total log events	40 552 429	08
Closed positions	1 342 059	02
Account records	455 752	01
Unique wallets (from logs)	447 599	08
Unique position keys	1 487 200	08
Daily ranking snapshots	261	03
Data span	491 days	08, 10
First event	2025-02-19	08
Latest event	2026-06-26	08

2.1 Platform and Chain Coverage

Platform concentration is moderate: Hyperliquid accounts for 53.6% of log events and Jupiter for 39.7%, followed by GMX-v2 (5.1%), APX Finance (1.0%), and Myx Finance (0.6%). The chain distribution mirrors platforms directly (Hyperliquid L1, Solana, Arbitrum, BSC). This concentration has implications for generalisability: results are primarily driven by two venues.

2.2 Asset Distribution

SOL (35.9%), BTC (27.2%), and ETH (21.6%) dominate event volume. The long tail includes HYPE, XRP, DOGE, ZEC, and meme tokens (FARTCOIN, PUMP). Critically, 76.7% of wallets trade only a single asset, which constrains cross-asset skill analysis to a small subsample.

2.3 Action Breakdown

Table 2: Log event distribution by action type.

Action	Count	Proportion
Open	21 594 000	53.2%
Close	17 339 702	42.8%
Liquidate	889 105	2.2%
Deposit	429 898	1.1%
Withdraw	299 724	0.7%

The Open/Close ratio is 0.803, reflecting the large number of still-open positions in the data. The liquidation rate is 4.12% of all opens, a non-trivial risk signal.

3 Population Characterisation

3.1 PnL Distribution

The aggregate PnL distribution confirms the extreme skew typical of crypto markets.

Table 3: PnL and ROI statistics (sample of 100 000 accounts).

Statistic	PnL (USD)	ROI (%)
Mean	\$-131.20	+283.24%
P10	\$-1 476.86	-5 059.89%
P25	\$-114.05	—
Median	\$-0.13	+0.00%
P75	\$+60.38	—
P90	\$+1 037.30	+2 176.30%
P99	\$+43 294.03	+21 558.26%

Only 43.6% of accounts are profitable. The median PnL is near zero (-\$0.13), consistent with a zero-sum market where most traders break even or lose after fees. The extreme ROI values (P99 > 200 $\times$) are driven by a small number of high-leverage trades and should be interpreted cautiously.

3.2 Win Rate Distribution

The mean win rate is 45.5% with a median of 50.0%. Notably, 29.6% of wallets have a win rate below 20% (including many with only one losing trade), while 21.4% have a win rate above 80% (likely one-trade winners). This bimodal shape at the extremes reflects small-sample noise rather than genuine skill differences.

3.3 Closed Trade Count Distribution

Table 4: Accounts meeting various closed-trade thresholds.

Threshold	N wallets	Proportion
≥ 1 trade	92 962	93.0%
≥ 5 trades	44 313	44.3%
≥ 10 trades	30 432	30.4%
≥ 20 trades	19 633	19.6%
≥ 50 trades	9 605	9.6%
≥ 100 trades	5 034	5.0%

The median wallet has only 3 closed trades. This extreme right-skew confirms that the vast majority of wallets lack sufficient history for reliable skill estimation without shrinkage.

4 Bayesian Feasibility

Script 04 assessed how many wallets provide enough information for posterior skill estimation to escape the prior.

At the target threshold of ≥ 20 trades, the posterior receives only 17% of its weight from the data (83% from the prior) under standard assumptions (prior SD = 0.10, observation SD = 0.20). This is precisely the scenario where Bayesian hierarchical modelling is most valuable—without it, all 20-trade wallets would be indistinguishable from noise. The credible interval width at 20 trades is ± 0.160 ROI, which may be too wide for individual wallet ranking but acceptable for population-level inference and top-decile identification.

Table 5: Trade sufficiency thresholds with Bayesian shrinkage simulation.

Threshold	Wallets	% of total	Shrinkage
≥ 5	202 170	44.4%	0.444
≥ 10	138 829	30.5%	0.286
≥ 20	89 067	19.5%	0.167
≥ 50	43 549	9.6%	0.074
≥ 100	22 886	5.0%	0.038

Table 6: Wallets meeting ≥ 20 trades threshold by platform.

Platform	Total wallets	≥ 20	%
hyperliquid	239 625	43 161	18.0%
jupiter	174 231	39 318	22.6%
gmx-v2	21 649	5 023	23.2%
apx-finance	19 673	1 458	7.4%
myx-finance	574	107	18.6%

4.1 Platform-Specific Suitability

Hyperliquid and Jupiter together provide 82 479 wallets with ≥ 20 trades, forming a solid population for Bayesian modelling. APX Finance’s low ratio (7.4%) reflects many one-time accounts.

5 Research Question 1: Side Asymmetry

Script 05 tested the core hypothesis of the thesis: are Long and Short trading skills separable?

5.1 Global Win Rates by Side

Aggregate win rates differ marginally: Long 48.49% (n=178 619) vs. Short 47.26% (n=121 381). The 1.23 percentage-point gap is small but directionally consistent with the known structural short-vol premium in perpetuals (funding rates).

5.2 Wallet-Level Orthogonality Test

For wallets with at least 5 trades on each side (n=748):

- **Pearson r (Long WR vs. Short WR): +0.336**
- **Pearson r (Long avg PnL vs. Short avg PnL): -0.233**

The correlation of +0.336 indicates **weak to moderate positive association**—Long and Short skill share some common component (general trading ability) but are far from perfectly correlated ($r < 0.5$). This supports the decomposition hypothesis: a single score conflates two distinct dimensions of skill.

The negative correlation between Long and Short PnL (-0.233) is a striking finding: wallets that profit on Longs tend to lose on Shorts, and vice versa. This suggests specialisation rather than general skill, further motivating side-aware scoring.

5.3 Skill Quadrant Analysis

The asymmetric quadrants (Q2 + Q4) comprise 34.9% of wallets—one in three wallets shows asymmetric skill. This is a substantial fraction and confirms that side-aware decomposition adds

Table 7: Skill quadrants at median Long WR (42.86%) and Short WR (43.99%).

Quadrant	N wallets	%
Q1: Good Long + Good Short	252	33.7%
Q2: Poor Long + Good Short	122	16.3%
Q3: Poor Long + Poor Short	235	31.4%
Q4: Good Long + Poor Short	139	18.6%

information beyond a single cross-side score.

5.4 Specialist Wallets

Wallets with extreme asymmetry exist: Long specialists (e.g., 100% Long WR vs. 20% Short WR) and Short specialists (e.g., 33% Long WR vs. 100% Short WR) appear in the data. These tail cases illustrate the practical value of side-aware scores for copy-trading decisions—copying a Long specialist on a Short trade would be value-destructive.

6 Leverage and Sizing Skill

6.1 Leverage vs. Outcome

Table 8: Win rate and average PnL by leverage bucket.

Leverage	N	Win%	Avg ROI	Avg PnL
<2x	5 683	47.8%	+0.68%	\$+9.69
2–5x	13 924	51.7%	+3.03%	\$+90.48
5–10x	14 895	51.1%	-0.48%	\$+5.30
10–20x	35 461	55.3%	+53.25%	\$+1 899.63
20–50x	71 970	49.6%	+7.46%	\$+425.34
>50x	26 259	48.1%	-160.68%	\$-258.17

The peak win rate at 10–20x leverage (55.3%) with substantially positive ROI (+53.25%) is noteworthy but may reflect selection effects rather than causal leverage skill. The dramatic negative ROI above 50x (−160.68%) confirms that extreme leverage destroys value on average, consistent with leverage being a risk-preference signal rather than a skill amplifier.

6.2 Sizing Skill (Van Loon 2018)

Measuring sizing skill as the difference in win rate between a wallet’s large and small trades:

- Positive sizing skill: 40.5% of wallets (n=2 902 with ≥ 10 trades)
- Negative sizing skill: 48.0%
- Mean sizing skill: −0.023 (near zero)
- Median: +0.000

The population-level distribution of sizing skill is centred at zero with heavy tails. 21.6% of wallets show strong negative sizing skill (< -0.2), suggesting systematic averaging-down behaviour (increasing position size on losing trades). Only 15.4% show strong positive sizing skill ($> +0.2$). These findings validate sizing skill as a distinct dimension worth modelling, though its signal-to-noise ratio is lower than side-level win rates.

7 Cross-Asset Skill Transfer

Cross-asset skill analysis was limited by single-asset dominance (76.7% of wallets). For the multi-asset subsample:

Table 9: Cross-asset win-rate Pearson correlations (wallets with ≥ 5 trades in both assets).

Asset A	Asset B	N wallets	r
BTC	ETH	90	+0.562
SOL	ETH	17	+0.707
SOL	XRP	8	+0.554
ETH	XRP	7	+0.891
BTC	SOL	27	+0.375
BTC	XRP	9	+0.477

All observed correlations exceed +0.3, with most above +0.5. This suggests that trading skill **does transfer across assets**—a wallet that wins on BTC also tends to win on ETH. However, the small sample sizes (n=7–90 wallets) imply wide confidence intervals. The generalisability of this finding is limited by the rarity of multi-asset wallets.

8 Baseline Critique

The existing baseline score (Script 03) reveals several design issues that motivate the proposed approach.

8.1 Score Compression

Table 10: Distribution of the existing composite score (latest snapshot, n=20 traders).

Statistic	Score	Range
Minimum	0.9688	
P25	0.9722	
Median	0.9773	
P75	0.9816	
Maximum	0.9881	
Range	0.0193	(0.969 to 0.988)

The score occupies only 1.9% of the $[0, 1]$ range. This compression makes it practically impossible to distinguish wallet quality—the difference between rank 1 and rank 20 is less than 0.02 points. The issue arises because all four component ratios are large positive numbers (e.g., Risk/Reward mean 57.1, WL Holding Time mean 207.8) that are then equally weighted and scaled arbitrarily.

8.2 Component Correlation

WL Holding Time and WL ROI Ratio are substantially correlated ($r = 0.652$), meaning they provide overlapping information. The equal-weighting scheme double-counts this shared variance. Risk/Reward is largely orthogonal to other components ($r = -0.038$ to $+0.342$), suggesting it captures a genuinely distinct dimension—but its contribution is diluted by the equal-weight scheme.

Table 11: Baseline component correlation matrix.

	Risk/Reward	WL Hold Time	WL ROI	Win%
Risk/Reward	1.000	−0.038	+0.342	+0.267
WL Hold Time		1.000	+0.652	+0.014
WL ROI Ratio			1.000	−0.208
Winning %				1.000

9 Pipeline: Feature Engineering and Walk-Forward Validation

The feature engineering pipeline (Stages 1–6) processes the raw log data through position reconstruction, per-wallet skill feature computation, label fetching from `closed_positions`, and out-of-sample validation. Results are stored as Parquet files under `data/processed/`.

9.1 Stage 1–4: Position Reconstruction and Baseline Computation

Stage 1 reconstructs completed positions from `logs` by matching Open events with their corresponding Close/Liquidate events via `positionKey`. Stage 2 computes per-wallet skill features using `SkillComputerService`, filtering to wallets with ≥ 5 trades. Stage 3 fetches ground-truth `realizedPnl` labels from `closed_positions`. Stage 4 computes baselines B1–B5 from the `accounts` and `daily_trader_rankings` collections.

Table 12: Pipeline data sizes after each stage.

Stage	Rows / Wallets	File
01 Reconstructed positions	1.34M positions	<code>positions.parquet</code>
02 Wallet features (≥ 5 trades)	57 013 wallets	<code>wallet_features.parquet</code>
03 Wallet labels (<code>closed_positions</code>)	424 899 wallets	<code>labels.parquet</code>
04 Baselines B1–B5	455 752 wallets	<code>baselines.parquet</code>

9.2 Stage 5: In-Sample Feature-Label Correlations

Pipeline 05 joins features with labels to identify which skill dimensions correlate with actual trading outcomes. The results confirm moderate predictive signal but reveal a critical discrepancy from the EDA-level analysis.

Table 13: Pipeline feature correlations with realized win rate and total PnL (in-sample).

Feature	r vs. <code>win_rate</code>	r vs. <code>total_PnL</code>
<code>overall_win_rate</code>	+0.434	+0.019
<code>long_win_rate</code>	+0.256	+0.013
<code>short_win_rate</code>	+0.204	+0.005
<code>mean_leverage</code>	−0.163	−0.002
<code>max_leverage</code>	−0.164	−0.002
<code>liquidation_rate</code>	−0.074	−0.002
<code>sizing_skill</code>	−0.022	+0.002
<code>side_asymmetry</code>	+0.010	+0.004
<code>avg_roi</code>	+0.006	+0.049
<code>leverage_adj_roi</code>	+0.006	+0.063

Several observations emerge:

- **overall_win_rate** ($r = +0.434$) is the strongest single predictor of in-sample realized win rate, confirming that past win rate has persistent signal.
- **Long and Short win rates** contribute separately but with overlapping information ($r = +0.256$ vs. $+0.204$).
- **Leverage measures are negatively correlated** with win rate: wallets using higher leverage tend to have lower win rates, consistent with leverage-as-aggression rather than leverage-as-skill.
- **leverage_adj_roi** has near-zero correlation with win rate but a modest $+0.063$ with total PnL—larger absolute PnL from leveraged positions.
- **sizing_skill** and **side_asymmetry** show near-zero correlation, suggesting these features capture orthogonal information not reflected in aggregate win rate.

9.2.1 Data-Quality Note: B1/B5 Baseline Comparison is Broken

The raw output of Stage 5 (`pipeline/outputs/05_feature_analysis/results.txt`) reports `b1_composite` and `b5_sharpe` correlations as `+nan` against both labels, while B2–B4 (raw PnL, ROI, win-rate) compute normally. The cause traces back to Stage 4: `daily_trader_rankings` only stores the *top-20* traders per daily snapshot (`total_traders_analyzed=20116` vs. `traders` list length 20, confirmed in Script 03’s own output), so `b1_composite` has non-null coverage for only 20 of 455752 wallets — after the inner join with the ≥ 5 -trade feature set, fewer than 5 rows survive and the correlation is undefined by Stage 5’s own guard (`len(sub) < 5 → nan`). `b5_sharpe` has much better raw coverage (107212 wallets, Stage 4), so its `nan` is not yet root-caused and should be checked directly (most likely a wallet-key mismatch or a degenerate/constant column after the join) before it is used in any downstream model. **This is a real limitation, not a display artifact:** B1 cannot currently serve as a fair full-population baseline, and any claim that the proposed score “beats B1” should be scoped to the 20 wallets B1 actually ranks, or B1 should be recomputed from the full `daily_trader_rankings` history (261 snapshots $\times$ 20 traders $\approx$ several thousand wallet-days) rather than the latest snapshot alone.

9.2.2 Critical Finding: Pipeline vs. EDA Disagreement on Side Correlation

The pipeline estimates **Pearson** $r(\text{long_WR}, \text{short_WR}) = -0.345$, while the EDA script 05 estimated $+0.336$ —the signs differ. This discrepancy arises from different data sources and filters:

- **EDA script 05** uses `closed_positions` directly (realized PnL per position) and requires ≥ 5 trades *per side* ($n=748$ wallets).
- **Pipeline** uses reconstructed positions from `logs` with ≥ 5 total trades ($n=57013$ wallets), where each wallet may have unbalanced Long/Short counts.
- The $+0.336$ estimate from script 05 measures the correlation *among wallets active on both sides*. The -0.345 pipeline estimate captures a broader population where Long and Short specialisation produces a negative association.

Both are informative: there is a weakly positive correlation for balanced traders (general skill) but a negative correlation at the population level (specialisation). This strengthens the case for side-aware modelling but demands careful specification of the target population.

9.2.3 Quadrant Analysis Confirms Asymmetry Dominance

The pipeline quadrant analysis ($n=53937$ wallets with non-null Long/Short WR), thresholding at $> 50\%$ WR for each side:

Table 14: Side-asymmetry quadrants from pipeline features.

Quadrant	N wallets	%
Q1: Dual skilled (Long > 50% & Short > 50%)	5 364	9.9%
Q2: Long specialist (Long > 50%, Short ≤ 50%)	15 049	27.9%
Q3: Short specialist (Long ≤ 50%, Short > 50%)	18 012	33.4%
Q4: Unskilled (both ≤ 50%)	15 512	28.8%

Only 9.9% of wallets are skilled on both sides. The majority (61.3%) are asymmetric—skilled on exactly one side. This is a substantially stronger signal of asymmetry than the EDA script 05 estimate (34.9% asymmetric), likely due to the larger wallet sample and different thresholding. It provides compelling evidence that a side-agnostic score misrepresents the majority of wallets.

9.3 Stage 6: Walk-Forward Validation (The Critical Test)

Pipeline 06 addresses the fundamental limitation of in-sample correlation analysis: past performance can trivially correlate with itself. By holding out the last 45 days of closed positions as an out-of-sample test window, we measure whether train-window features predict *future* wallet performance.

Table 15: Walk-forward validation: out-of-sample feature correlations.

Feature	r vs. future WR	r vs. future PnL
overall_win_rate*	+0.110	+0.006
long_win_rate	+0.054	+0.030
avg_roi*	+0.057	+0.003
short_win_rate	+0.037	+0.001
leverage_adj_roi	+0.055	+0.003
side_asymmetry	+0.012	+0.018
mean_leverage	−0.101	+0.001
liquidation_rate	−0.108	−0.001
sizing_skill	−0.050	+0.009
n_trades	−0.086	−0.002
total_pnl*	−0.039	+0.102

*Naive baseline (computed identically on the train window).

Only 1 231 wallets have sufficient trades (≥ 5 in both windows) for this test—a dramatic reduction from the 57 013 in the in-sample analysis. The out-of-sample correlations are substantially weaker than in-sample, with three critical implications:

Signal collapse: The best predictor (overall_win_rate) drops from $r = +0.434$ in-sample to $r = +0.110$ out-of-sample—a 75% signal decay. Most of the apparent in-sample signal comes from look-ahead structure or overfitting, not genuine skill persistence.

Risk signals are the most stable predictors: liquidation_rate (−0.108) and mean_leverage (−0.101) are the strongest directional signals out-of-sample, exceeding all win-rate-based features except overall_win_rate. This suggests that **what a wallet does wrong (excessive risk)** is more predictive than **what it does right**.

Feature orthogonality is confirmed: The near-zero correlations of sizing_skill, side_asymmetry, and trade_frequency with future WR confirm they capture orthogonal information—but this

information has minimal predictive power at the population level in a 45-day window.

10 Bayesian Skill Model (Stage 7)

Pipeline Stage 7 implements the proposed Bayesian hierarchical model from the research proposal (§3.2), using an empirical-Bayes Normal-Normal conjugate model (`src/skill_model/empirical_bayes.py`). Following a code update implementing NS4/NS5 (§“Next Steps”), the model now covers five dimensions $d \in \{\text{buy, sell, timing, sizing, liquidation}\}$ instead of three. BUY/SELL/TIMING/LIQUIDATION are rate-style (binomial-variance) dimensions:

$$\theta_w^{(d)} \sim \mathcal{N}(\mu^{(d)}, \tau^{(d)2}), \quad \hat{s}_w^{(d)} \mid \theta_w^{(d)} \sim \mathcal{N}\left(\theta_w^{(d)}, \frac{\sigma_w^{(d)2}}{N_w}\right),$$

where $\mu^{(d)}$ and $\tau^{(d)2}$ are estimated by method of moments (Morris, 1983) and the posterior mean is a convex combination of the wallet’s raw estimate and the population mean, weighted by the shrinkage factor $\lambda_w = \sigma_w^2 / (\sigma_w^2 + \tau^2)$. The new SIZING dimension uses `fit_continuous_dimension` on $\log(W/L)$ (Van Loon 2018’s win/loss ratio, in log space so the Normal likelihood is appropriate), with sampling variance from the delta method: $\widehat{\text{Var}}[\log(W/L)] \approx \text{Var}(r^{win}) / (n_{win} \bar{r}_{win}^2) + \text{Var}(r^{loss}) / (n_{loss} \bar{r}_{loss}^2)$.

Table 16: Empirical-Bayes hyperparameters for each skill dimension (updated: 5 dimensions).

Dimension	N wallets	μ	τ^2	Mean shrinkage
BUY (long_win_rate)	16 659	0.4689	0.03479	0.386
SELL (short_win_rate)	8 429	0.4777	0.03145	0.405
TIMING (overall_win_rate)	48 660	0.4823	0.00610	0.786
LIQUIDATION (survival_rate)	48 660	0.1888	10.26628	0.001
SIZING (log W/L ratio)	39 600	0.1066	1.13820	0.325

The combined `bayes_score` is now an equal-weighted average of the z-scored posterior means of buy/sell/timing/sizing (liquidation is excluded from this average and used separately, see below). Two findings stand out. **First, TIMING still shrinks heavily (0.786)**—consistent with the earlier discussion, because the train-window trade-count constraint is shorter than the full-history 20-trade threshold assumed in the feasibility analysis (Table 5). **Second, LIQUIDATION shrinks almost not at all (0.001)**: its population mean survival rate of 0.189 (i.e. an 81.1% average liquidation rate, matching the B6 population mean in Table 20) reflects that this data set’s wallets are liquidated on the large majority of trades, and because n is typically large for this ratio, sampling noise is small relative to the very large between-wallet variance ($\tau^2 = 10.27$)—almost every wallet’s raw survival rate is trusted almost as-is.

A new `van_loon_score` is also computed as the literal multiplicative form of Eq. 1.2: `posterior_timing` × `exp(posterior_sizing)` × $\sqrt{n_{\text{trades}}}$ − (1 − `posterior_liquidation`), directly operationalising the Van Loon (2018) decomposition from §1 rather than an equal-weighted z-score average.

11 Baseline Comparison (Stage 8)

Pipeline Stage 8 compares the proposed `bayes_score` (now 4-dimensional: buy/sell/timing/sizing) and the new `van_loon_score` against baselines B1–B5 using Spearman rank correlation (robust to outliers and non-linearity) and top-decile hit rate.

Adding sizing and liquidation as explicit dimensions overturns the previous ranking:

Table 17: Out-of-sample baseline comparison: Spearman rank correlation and top-decile hit rate (updated with sizing/liquidation dimensions).

Score	r_s vs. future WR	r_s vs. future PnL	Hit rate
van_loon_score	+0.068	+0.022	40.7%
bayes_score	+0.058	+0.011	42.6%
posterior_buy	+0.084	+0.040	41.0%
posterior_sell	+0.145	+0.027	45.9%
posterior_timing	+0.134	+0.060	46.9%
posterior_sizing	-0.058	-0.013	36.9%
posterior_liquidation	+0.228	+0.094	44.9%
b1_composite	+NAN	+NAN	NAN%
b2_pnl	+0.106	+0.030	47.8%
b3_roi	+0.142	+0.051	52.2%
b4_win_rate	+0.072	-0.029	40.4%
b5_sharpe	+0.075	-0.022	37.5%

- **posterior_liquidation** ($r_s = +0.228$) is now the single strongest score in the entire comparison, beating even b3_roi (+0.142). This confirms the NS5 hypothesis directly: liquidation avoidance, treated as its own shrunk posterior rather than a raw correlate (Pipeline 06 found $r = -0.108$ for liquidation_rate), is the most informative signal available in this data set.
- **posterior_sizing** ($r_s = -0.058$) is actively *negative*—wallets with a shrunk higher win/loss ratio in the train window tend to do *worse* out-of-sample. This is the opposite of the Van Loon (2018) prediction and is consistent with the population-level finding (§ Leverage and Sizing Skill) that sizing skill is centred at zero with heavy, noisy tails; the continuous-dimension shrinkage evidently does not clean this up enough over a 45-day window.
- **Both composite scores now underperform their best single dimension:** bayes_score (+0.058, down from +0.109 in the 3-dimension model) is dragged down by including the negative-signal posterior_sizing in the equal-weighted average, and van_loon_score (+0.068) inherits the same problem because it multiplies by $\exp(\text{posterior_sizing})$.
- **posterior_sell** ($r_s = +0.145$) and **posterior_timing** ($r_s = +0.134$) remain strong and unchanged, still consistent with Lim et al. (2022) and Van Loon (2018) respectively.
- **b1_composite is NAN:** the existing baseline covers only the top-20 wallets per daily snapshot, making a full-population comparison impossible.

The practical implication is that naïve equal-weighting across dimensions is actively harmful once a noisy or inverted-sign dimension (sizing) is included—composite scores should be built from a learned weighting (NS3) that can down-weight or drop posterior_sizing, rather than an unweighted average.

12 Luck vs. Skill Validity (Stage 9)

Stage 9 implements two complementary null-hypothesis tests drawn from the literature review.

12.1 Exact Binomial Test (Kosowski et al. 2006)

Following Kosowski et al. (2006), we test each wallet’s train-window hit rate against the population base rate (0.4435) using an exact binomial test. Wallets with a significantly better hit rate

(one-sided $p < 0.05$) are flagged as potential skilled traders:

- Wallets significant at $p < 0.05$: **84 of 1,231 (6.82%)**
- Reference: Gomez-Cram et al. (2026) flag 3.14% of Polymarket accounts via an analogous sign-randomisation persistence test
- **58.5% of bayes_score top-decile wallets** are also binomial-significant—the Bayesian ranking partially but imperfectly aligns with the classical frequentist test

12.2 Monte-Carlo Random-Allocation Null (Papaioannou et al. 2024)

Following the M6 Investment Challenge methodology (Papaioannou et al. 2024, [arXiv:2412.04490](#)), we simulate 10,000 chance-trader populations with the same trade-count distribution as the real data but random win/loss outcomes at the population base rate. The real top-decile mean *future* win rate (out-of-sample) is compared against the null distribution of *train*-window top-decile means from chance traders:

- Real top-decile mean future WR: **0.5010**
- Null (chance-trader) top-decile mean: **0.7274 $\pm$ 0.0108**
- **The real value falls at the 0.0th percentile of the null.**
- Conclusion: **Cannot reject the null hypothesis** that top-decile out-of-sample performance is explainable by chance.

This is a sobering result: even the best wallets in our scoring system do not outperform a random-allocation benchmark in the 45-day holdout window. The null’s high mean (0.7274) is an artifact of the noise—when pick-5-trades-at-random from a 44.4% base-rate population, the top 10% of 1,231 simulated wallets will have high win rates by construction (order statistics). This does not invalidate the relative ranking produced by `bayes_score`—it merely confirms that the absolute level of skill in the population is low and that 45-day labels are too noisy for fine-grained discrimination.

13 Additional Baselines

13.1 B7: Sign-Randomisation Skill Classifier (Stage 10)

We reproduce the Gomez-Cram et al. (2026) “skilled winner” classifier (synthesised in Nechepurenko 2026, [arXiv:2605.02287](#)) using an exact binomial test of each wallet’s directional-accuracy persistence against the population base rate. The original code and labels are not publicly released, so our implementation follows the paper’s method description.

Table 18: B7 sign-randomisation results compared with Gomez-Cram et al. (2026).

Metric	This study	Gomez-Cram et al. (2026)
Population base rate	46.4%	—
Wallets flagged ($p < 0.05$)	1,634 (3.36%)	3.14%

The flagged proportion (3.36% vs. 3.14%) is remarkably consistent across domains (perp futures vs. prediction markets), suggesting the same fraction of wallets exhibit persistent directional accuracy regardless of market structure.

13.2 B8: Wash-Trade / Bot Pre-Filter (Stage 11)

We adapt the graph-based wash-trade heuristic of Ashfaq (2023, [arXiv:2305.01543](#))—originally designed for NFT collusion detection—to perpetual futures positions. Since our data has no

counterparty field (positions are wallet-vs.-protocol), the original cycle-detection approach does not transfer directly. Instead, we flag wallets whose median position duration is below 1 hour with at least 20 trades, indicating rapid open-close cycles consistent with bot or wash-trading activity.

Table 19: B8 wash-trade / bot detection results.

Metric	This study (perp)	Ashfaq (2023, NFT)
Wallets evaluated	57,013	—
Flagged	1,079 (1.89%)	0.11%

The higher flag rate (1.89% vs. 0.11%) reflects the adapted rule (duration-based rather than cycle-based) and the inherently shorter holding periods in perp futures vs. NFT markets. After intersecting with the 1,348-wallet walk-forward sample, only 29 wallets (2.15%) are excluded by the B8 filter.

13.3 B6: zScore Reimplementation (Stage 12)

We reimplement the zScore framework (Anon et al. 2025, [arXiv:2507.20494](#)) using rule-based “blueprint” features adapted to perp position data. The original uses a deep residual neural network on Uniswap v3 LP + swap behaviour; our reimplementation is scope-adapted (no LP side, no Deposit/Withdraw signals) and uses a heuristic composite of volume, trade frequency, holding time, and liquidation rate rather than a trained neural network.

Table 20: B6 zScore reimplementation blueprint component means.

Component	Population mean
Volume (USD)	525,603.52
Frequency (trades/day)	0.96
Holding time (minutes)	780.17
Liquidation rate	0.81

The zScore is computed as a standardised composite of these four components, inverted so that higher values indicate better behaviour (lower liquidation rate, higher volume, etc.).

14 Full Baseline Comparison (Stage 13)

Pipeline Stage 13 consolidates all eight baselines (B1–B8) plus the proposed `bayes_score` and the new `van_loon_score` into a single out-of-sample comparison.

The following key findings emerge:

1. **b3_roi (raw ROI) is still the single best *aggregate* baseline**, achieving $r_s = +0.142$ and a 52.2% top-decile hit rate. ROI captures PnL magnitude, but is now beaten by one individual Bayesian posterior: `posterior_liquidation` ($r_s = +0.228$, Table 17), which this table does not show because it only lists the combined `bayes_score/van_loon_score`, not the individual dimensions.
2. **Both proposed composite scores regressed relative to the previous (3-dimension) model: `bayes_score` fell from $r_s = +0.109$ to $+0.058$, and the new `van_loon_score` lands at $+0.068$ —both worse than `b2_pnl`, `b3_roi`, `b6_zscore`, and `b7_score`.** The cause is directly traceable to §“Baseline Comparison”: the newly-added `posterior_sizing` dimension carries a *negative* out-of-sample correlation (-0.058) and drags down any composite that averages or multiplies it in.

Table 21: Full baseline comparison: all scores vs. out-of-sample future performance (updated with `van_loon_score`, 4-dim `bayes_score`).

Score	r_s WR	r_s PnL	Hit rate	Note
<code>van_loon_score</code>	+0.068	+0.022	40.7%	
<code>bayes_score</code>	+0.058	+0.011	42.6%	
<code>b1_composite</code>	NAN	NAN	NAN%	Only 20 wallets/snapshot
<code>b2_pnl</code>	+0.106	+0.030	47.8%	Full history
<code>b3_roi</code>	+0.142	+0.051	52.2%	Full history
<code>b4_win_rate</code>	+0.072	−0.029	40.4%	Full history
<code>b5_sharpe</code>	+0.075	−0.022	37.5%	Full history
<code>b6_zscore</code>	+0.125	+0.015	32.4%	Train window only
<code>b7_score</code>	+0.131	+0.058	46.3%	Train window only

3. **`b7_score` (sign-randomisation) remains the second-best composite baseline** at $r_s = +0.131$, validating the Gomez-Cram et al. (2026) methodology on perp data.
4. **`b6_zscore` ($r_s = +0.125$) remains competitive** despite being a simple heuristic composite rather than the original neural network.
5. **`b4_win_rate` and `b5_sharpe` show negative correlation with future PnL** (−0.029 and −0.022 respectively), meaning wallets with higher past win rates and Sharpe ratios tend to underperform on future absolute PnL—a reversal effect.

14.1 Robustness Check: Excluding B8-Flagged Wallets

After removing the 29 wallets flagged as likely wash/bot traders (2.15% of the sample), the ranking shifts slightly:

Table 22: Robustness check: full comparison with B8-flagged wallets excluded.

Score	r_s WR	r_s PnL	Hit rate	Note
<code>van_loon_score</code>	+0.093	+0.023	42.9%	
<code>bayes_score</code>	+0.045	+0.011	42.1%	
<code>b1_composite</code>	NAN	NAN	NAN%	
<code>b2_pnl</code>	+0.107	+0.030	47.4%	
<code>b3_roi</code>	+0.146	+0.052	54.1%	
<code>b4_win_rate</code>	+0.061	−0.028	41.4%	
<code>b5_sharpe</code>	+0.078	−0.022	37.8%	
<code>b6_zscore</code>	+0.155	+0.015	38.3%	
<code>b7_score</code>	+0.113	+0.059	45.9%	

The B8 filter *improves* `b3_roi` and `b6_zscore` (bots with extreme ROI or anomalous zScore components are removed) and now also *improves* `van_loon_score` (+0.068 → +0.093), while *hurting* `bayes_score` further (+0.058 → +0.045). The diverging direction between the two proposed scores is itself informative: `van_loon_score`’s $\sqrt{n_{\text{trades}}}$ breadth term rewards wallets with more (genuine, non-bot) trades once wash-traders are removed, whereas `bayes_score`’s unweighted z-score average has no such term and continues to be dragged down by `posterior_sizing` regardless of the B8 filter.

15 Label Reliability and a Learned Ranker (Stage 14)

Every out-of-sample number reported up to this point ($r_s \approx 0.06$ – 0.23) is evaluated on the same population: wallets with ≥ 5 trades in the 45-day holdout window. This section shows that this population is the wrong one—its label is almost pure noise—and that once this is corrected, a *learned* combination of the Bayesian posteriors decisively beats raw ROI and the equal-weighted composites.

15.1 Data-Quality Fix: Degenerate-Collateral Rows

An audit of `positions.parquet` found the `roi` column spanning $[-3.9 \times 10^{13}, 1.2 \times 10^{14}]$, driven by 83 positions with `entry_collateral_usd` ≤ 0 (division by non-positive collateral). Because the SIZING dimension is the log-ratio of *mean* winning to *mean* losing ROI, a single such row corrupts an entire wallet’s sizing estimate. `SkillComputerService` now drops these rows and winsorizes ROI to its $[0.1\%, 99.9\%]$ quantiles before any aggregation. This lowered the SIZING between-wallet variance ($\tau^2 : 1.138 \rightarrow 0.867$) but—importantly—**did not flip** `posterior_sizing`’s out-of-sample sign ($r_s = -0.062$ after cleaning). The negative sizing signal is therefore a genuine mean-reversion effect, not an outlier artifact: wallets with a high train-window win/loss ratio tend to regress *below* average out-of-sample.

15.2 The Label Is Dominated by Sampling Noise

For a wallet with n future trades and true win probability p , the observed `future_win_rate` has binomial sampling variance $p(1-p)/n$. Define the label *reliability* as the fraction of observed variance that is not sampling noise:

$$\rho = \frac{\max(\text{Var}(\text{future_WR}) - \mathbb{E}[p(1-p)/n], 0)}{\text{Var}(\text{future_WR})}.$$

On the ≥ 5 -future-trade population (the evaluation set used in Stages 8–13), $\rho = 0.013$: the observed label standard deviation (0.183) is $1.007\times$ the standard deviation expected under *pure* coin-flipping. **The label is statistically indistinguishable from luck.** This is the mechanistic explanation for the Stage 9 Monte-Carlo null result—top-decile performance cannot be distinguished from chance because, on this population, there is essentially no persistent between-wallet signal in the label to detect. Requiring more future trades restores signal:

Table 23: Label reliability ρ of `future_win_rate` by minimum future-trade threshold (Stage 14, post data-cleaning population).

Min future trades	N wallets	Reliability ρ
≥ 5 (Stages 8–13 eval set)	488	0.146
≥ 10	238	0.320
≥ 20	70	0.651

(The ≥ 5 reliability here is 0.146 rather than 0.013 because Stage 14 additionally requires all five posteriors to be non-null, which selects wallets with longer train histories and hence somewhat less noisy labels. Even so, 85% of the label variance at this threshold is noise.) The practical rule this establishes: **no Spearman/hit-rate number should be read without the reliability of the label it is measured against**, and comparisons on the ≥ 5 population are near-uninformative.

15.3 Learned Ranker (LambdaMART)

Stage 14 replaces the hand-specified combinations (`bayes_score`, `van_loon_score`) with a LightGBM LambdaMART ranker (`objective=lambdarank`) trained to rank wallets by future win rate,

using the five train-window posteriors plus n_{trades} as features. It optimises NDCG directly, so it can learn to down-weight the noisy/inverted sizing dimension rather than averaging it in. Performance is reported by 5-fold cross-validation (out-of-sample within the scored population) at each reliability regime, against the two strongest baselines from Stage 13 (b3_roi, full-history) and the equal-weighted `bayes_score`:

Table 24: Learned ranker vs. baselines, by label-reliability regime (Stage 14). Spearman is 5-fold cross-validated for the learned ranker; NDCG@10% is mean $\pm$ std across folds.

Min future trades (ρ)	N	learned_ranker	b3_roi	posterior_liq.	bayes_score
≥ 5 ($\rho=0.15$)	488	+0.205	+0.150	+0.305	+0.132
≥ 10 ($\rho=0.32$)	238	+0.304	+0.172	+0.422	+0.210
≥ 20 ($\rho=0.65$)	70	+0.385	+0.166	+0.394	+0.386

Three findings, in decreasing order of confidence:

1. **Signal is real and rises monotonically with label reliability.** As ρ climbs $0.15 \rightarrow 0.32 \rightarrow 0.65$, the learned ranker’s cross-validated r_s climbs $+0.205 \rightarrow +0.304 \rightarrow +0.385$. The earlier “everything looks like luck” conclusion was an artifact of evaluating on a noise-dominated label, *not* evidence that skill is unrecoverable. This is the central corrected result of the report.
2. **A learned combination beats both raw ROI and the equal-weighted composite at every reliability level.** The ranker exceeds b3_roi ($+0.205$ vs. $+0.150$ at ≥ 5 ; $+0.385$ vs. $+0.166$ at ≥ 20) and the equal-weighted `bayes_score`, directly confirming NS3: the problem was never the dimensions, it was averaging them with fixed equal weights.
3. **A single risk dimension remains a remarkably strong, simpler baseline.** `posterior_liquidation` alone matches or beats the seven-feature ranker at the lower thresholds ($+0.305$ vs. $+0.205$ at ≥ 5 ; $+0.422$ vs. $+0.304$ at ≥ 10) and only ties it at ≥ 20 ($+0.394$ vs. $+0.385$). This tempers the “learned model wins” story: much of the recoverable signal is a single, interpretable feature (liquidation avoidance), and the gain from a full learned ranker over that one feature is not established at ≥ 20 given $n = 70$ and a fold-to-fold std of ± 0.29 .

Caveats. The ≥ 20 regime rests on 70 wallets with wide cross-validation error bars (± 0.21 – 0.29), and the ranker uses train-window features only while b3_roi uses full history—so the learned ranker’s margin over ROI is if anything understated, but its margin over `posterior_liquidation` is within noise. Rolling multi-fold walk-forward (Stage 15, §16) supplies exactly this and is reported next. A separate data-integrity issue surfaced here: `baselines.parquet` contains $\sim 4,000$ duplicate wallet keys, which inflate the joined population in Stages 8/13 (e.g. $1,230 \rightarrow 1,347$); Stage 14 de-duplicates before joining, and Stage 4 should be corrected upstream.

16 Rolling Walk-Forward Validation (Stage 15)

Every out-of-sample number in Stages 6–14 rests on a *single* 45-day holdout—one draw from one market regime. Stage 15 replaces it with a rolling walk-forward: an expanding train window and five consecutive, non-overlapping 45-day test windows. Each fold independently recomputes all empirical-Bayes posteriors on its own train window (no leakage across the cutoff), evaluates every score against that fold’s future labels (wallets with ≥ 10 future trades, so the label reliability is meaningful—mean $\rho = 0.55$ across folds), and re-fits the LambdaMART ranker with within-fold cross-validation. Reporting mean $\pm$ std across folds converts the Stage 14 point estimates into interval estimates and tests regime sensitivity.

The multi-fold view materially revises the single-fold conclusions of Stages 8–14:

Table 25: Per-fold Spearman vs. future win rate (fold 0 = most recent test window). Each fold refits posteriors on its own expanding train window.

Fold	N	ρ	learned	post_liq	post_timing	post_sell	b3_roi
0	238	0.320	+0.290	+0.422	+0.273	+0.253	+0.172
1	121	0.604	+0.383	+0.290	+0.525	+0.559	+0.170
2	143	0.552	+0.473	+0.349	+0.522	+0.449	+0.183
3	191	0.606	+0.512	+0.359	+0.578	+0.490	+0.174
4	177	0.673	+0.426	+0.300	+0.522	+0.367	+0.166

Table 26: Walk-forward aggregate: mean $\pm$ std Spearman across the five folds.

Score	mean $r_s \pm \text{std}$
posterior_timing	+0.484 $\pm$ 0.108
posterior_sell	+0.424 $\pm$ 0.106
learned_ranker	+0.417 $\pm$ 0.077
bayes_score	+0.395 $\pm$ 0.097
posterior_liquidation	+0.344 $\pm$ 0.048
b3_roi	+0.173 $\pm$ 0.006

1. **Every Bayesian score decisively beats raw ROI.** b3_roi is remarkably *stable* ($+0.173 \pm 0.006$) but is the weakest score by a wide margin; all five posterior-based scores land at $r_s = 0.34\text{--}0.48$, a $2\text{--}3\times$ improvement. **The report’s original “the proposed model cannot beat ROI” headline was an artifact of a single noisy fold evaluated on a near-pure-noise label**, not a property of the method. This is the single most important correction in the report.
2. **The best single score is timing, not liquidation.** posterior_timing ($+0.484 \pm 0.108$, Van Loon hit-rate skill) is the strongest on average. posterior_liquidation’s Stage 14 dominance ($r_s = +0.42$) was *fold-0-specific* (its best fold); across regimes it is the *most stable* posterior (± 0.048) but only mid-ranked on mean ($+0.344$). Single-fold rankings were misleading.
3. **The learned ranker’s value is variance reduction, not peak lift.** learned_ranker ($+0.417 \pm 0.077$) does *not* exceed its best single component (timing, $+0.484$)—honestly, a learned combination does not beat simply using the timing posterior on this data. What it does buy is the lowest fold-to-fold volatility (± 0.077) among the strong scores, i.e. a more regime-robust ranking at a small cost in mean. Whether that trade is worthwhile is an application decision.
4. **Sell skill is confirmed strong** ($+0.424 \pm 0.106$), consistent with Lim et al. (2022); it and timing together dominate, exactly the two dimensions the side-aware / Van Loon decomposition was built to isolate.

Remaining caveats. Five folds still share overlapping expanding-train data and one asset universe; the std across folds ($\pm 0.05\text{--}0.11$) understates true regime uncertainty. Fold 0 (the Stage 14 holdout) has the lowest reliability ($\rho = 0.32$) and is the weakest fold for the win-rate-based scores, which is consistent with the reliability mechanism rather than an outlier. b3_roi retains its full-history information advantage yet still loses decisively, strengthening the conclusion.

16.1 Why the Proposed Bayesian Model Does Not Yet Beat ROI

The original finding of this report was that the 3-dimensional `bayes_score` ($r_s = +0.109$) underperformed raw ROI ranking ($r_s = +0.142$) out-of-sample, attributed to the model shrinking only *binary win rates* while ROI captures *PnL magnitude*. Following the code update that added the SIZING (log W/L ratio, Van Loon 2018) and LIQUIDATION (survival rate) dimensions (§ “Bayesian Skill Model”), the diagnosis needs revising:

- **Adding sizing did not fix the problem—it made the composite worse.** `bayes_score` fell to $r_s = +0.058$ once `posterior_sizing` (itself $r_s = -0.058$ out-of-sample) was folded into the equal-weighted average. The hypothesis that “PnL magnitude via sizing skill is the missing ingredient” is not supported by this implementation: the shrunk $\log(W/L)$ posterior does not persist out-of-sample over a 45-day window, even though it is a distinct dimension (Table 16 shows moderate shrinkage of 0.325, i.e. genuine signal survives the prior).
- **The liquidation dimension is the actual missing ingredient.** `posterior_liquidation` alone reaches $r_s = +0.228$, comfortably the best score in the entire study (Table 17), consistent with Pipeline 06’s finding that `liquidation_rate` was the most stable *raw* out-of-sample predictor. Risk avoidance, not PnL magnitude, is what best complements timing/hit-rate skill in this data set.
- **ROI (`b3_roi`, $r_s = +0.142$) is still the strongest *aggregate* baseline**, and both proposed composites (`bayes_score` $+0.058$, `van_loon_score` $+0.068$) remain below it—worse than before this update, not better.

The methodological lesson has two parts. First, the earlier “the proposed model cannot beat ROI” conclusion was a *measurement* failure, not a model failure: it was computed on a single fold against a label that is ≈ 85 – 99% sampling noise (§15). Under reliability-aware, multi-fold walk-forward evaluation (Stage 15, §16) **every Bayesian posterior score beats raw ROI by 2–3 $\times$** ($r_s \approx 0.34$ – 0.48 vs. 0.173). Second, the Van Loon (2018) decomposition is directionally correct—`posterior_timing` (hit-rate skill) and `posterior_sell` (exit skill) are the two strongest dimensions—but a *learned* combination of all five dimensions does not, on this data, exceed simply using the single timing posterior ($+0.417$ vs. $+0.484$ mean). The learned ranker’s benefit is lower fold-to-fold variance (± 0.077 vs. ± 0.108), i.e. regime robustness, not higher mean lift. The honest bottom line: the side-aware / timing decomposition is the win; the machinery to combine dimensions is a secondary, still-open question (NS3).

16.2 Relationship to the Literature

The results align with three key findings from the literature review:

- L1: Timing and sell skill dominate** (Van Loon, 2018; Lim et al., 2022): under multi-fold walk-forward (Stage 15), `posterior_timing` ($+0.484 \pm 0.108$) and `posterior_sell` ($+0.424 \pm 0.106$) are the two strongest scores, matching Van Loon’s emphasis on hit-rate (timing) skill and Lim et al.’s finding that exit (sell) decisions drive aggregate performance. (The single-fold Stage 8 estimate of $r_s = +0.145$ for sell was a noisy underestimate.)
- L2: Skilled-wallet proportion is 3%** (Gomez-Cram et al., 2026): Our B7 reproduction flags 3.36% of wallets, nearly identical to the 3.14% reported on Polymarket.
- L3: The “luck dominates” reading was a label-noise artifact** (cf. Papaioannou et al., 2024; Choi et al., 2025): the Monte-Carlo null could not reject chance *on the ≥ 5 -trade population*, but that label has reliability $\rho = 0.013$. Restricting to reliably-labelled wallets (≥ 10 future trades, $\rho \approx 0.55$) recovers strong, regime-stable skill signal (Stage 15). Heavy-tailed returns still make *absolute* top-decile performance hard to certify, but *relative* skill ranking is clearly recoverable.

16.3 Limitations

1. **Limited fold count (partly addressed):** Stages 6–14 used a single 45-day holdout; Stage 15 (§16) extends this to five non-overlapping folds and reports mean $\pm$ std, but the folds share overlapping expanding-train data and one asset universe, so the reported std (± 0.05 – 0.11) still understates true regime uncertainty. More folds and a purged/embargoed split are future work.
2. **Small per-fold sample:** Reliably-labelled populations are 121–238 wallets per fold (≥ 10 future trades). Rankings separated by less than the fold std ($\sim \pm 0.1$) are not meaningfully different.
3. **No crowd-adjustment (RQ3):** The current model scores wallets independently. If skill decays with follower count (Liu et al., 2023; Gao et al., 2026), scores should be discounted for crowded wallets—but the follower graph is not available in this database.
4. **Information advantage for B2–B5:** Baselines b2_pnl, b3_roi, b4_win_rate, and b5_sharpe use the full account history, while the Bayesian scores use only the train window. This gives the baselines a slight advantage that is not yet quantified.

17 Key Findings Summary

- F1: Large but right-skewed population.** 447 599 wallets exist, but 80.4% have fewer than 20 trades. Bayesian shrinkage is essential.
- F2: Side correlation is population-dependent.** For the broad pipeline population ($n=57\,013$), Long vs. Short WR $r = -0.345$ (specialisation). For balanced traders ($n=748$), $r = +0.336$ (general skill). Both support side-aware decomposition but target different populations.
- F3: Asymmetric skill is dominant.** Pipeline quadrant analysis shows 61.3% of wallets are asymmetric (skilled on exactly one side), far exceeding the 34.9% from the smaller EDA sample. Only 9.9% are dual-skilled.
- F4: Baseline score is compressed and partially redundant.** Existing scores occupy only 1.9% of the $[0, 1]$ range. Two of four components are highly correlated ($r = 0.652$).
- F5: Leverage optima exist.** Peak win rate at 10–20x leverage; extreme leverage ($> 50x$) destroys value. Leverage-adjusted skill scores would add information.
- F6: Sizing skill is weak at the population level.** Mean sizing skill is near zero (median $+0.000$), but 21.6% of wallets show strong negative sizing (averaging down), a distinct behavioural signal.
- F7: Cross-asset skill transfer is plausible but data-limited.** Correlations among multi-asset wallets exceed $+0.5$, but only 23.3% of wallets trade more than one asset.
- F8: Out-of-sample signal is weak.** Walk-forward validation (Pipeline 06) shows a 75% signal collapse from in-sample ($r = +0.434$) to out-of-sample ($r = +0.110$) for the best predictor. Risk signals (liquidation_rate -0.108 , mean_leverage -0.101) are the most stable directional predictors.
- F9: Bayesian model extended to 5 dimensions; composite still does not beat ROI, but one dimension now beats everything.** Adding SIZING and LIQUIDATION posteriors (§“Bayesian Skill Model”) dropped bayes_score further, from $r_s = +0.109$ to $+0.058$ (raw ROI, $r_s = +0.142$, is unchanged and still the best aggregate baseline), because posterior_sizing is out-of-sample *negative* ($r_s = -0.058$). However, **posterior_liquidation alone reaches $r_s = +0.228$** , the best score in the whole study—risk avoidance, not PnL

magnitude, is the actual missing ingredient. Posterior sell skill ($r_s = +0.145$) remains the strongest *directional* dimension, consistent with Lim et al. (2022).

- F10: Skilled-wallet proportion is $\sim 3\%$.** Both the binomial (6.82%) and sign-randomisation (3.36%) tests flag a small fraction of wallets as persistently skilled, consistent with Gomez-Cram et al. (2026) and Kosowski et al. (2006).
- F11: The “luck cannot be rejected” result was a label-noise artifact.** The Stage 9 Monte-Carlo null could not distinguish top-decile performance from chance only because the ≥ 5 -future-trade label has reliability $\rho = 0.013$ ($\approx 99\%$ sampling noise). On reliably-labelled wallets (≥ 10 future trades, $\rho \approx 0.55$) the signal is strong and stable—this is not evidence that skill is absent.
- F12: Under multi-fold walk-forward, every Bayesian score beats raw ROI by $2\text{--}3\times$.** Stage 15 (5 folds, expanding train): `posterior_timing` $+0.484 \pm 0.108$, `posterior_sell` $+0.424 \pm 0.106$, `learned_ranker` $+0.417 \pm 0.077$, `bayes_score` $+0.395 \pm 0.097$, `posterior_liquidation` $+0.344 \pm 0.048$, all vs. `b3_roi` $+0.173 \pm 0.006$. Timing (Van Loon hit-rate skill) is the best single score; the learned ranker is the most *stable* but not the highest-mean. The single-fold “ROI wins” finding of Stages 8–14 does not survive proper walk-forward.
- F13: A data-quality audit was necessary.** 83 positions with non-positive collateral produced ROI up to 10^{14} , corrupting mean-based features; `baselines.parquet` carries $\sim 4,000$ duplicate wallet keys inflating joined populations. Both are fixed/guarded in Stages 14–15 and flagged for upstream correction.
- F14: Wash-trade filter identifies 1.89% of wallets as likely bots.** Removing them improves ROI and zScore rankings but weakens the Bayesian score, suggesting the Bayesian model partially relied on non-genuine signals.

18 Next Steps

18.1 Immediate: Improve the Bayesian Model (Address P1 Findings)

- NS1: Multi-dimensional shrinkage:** Extend the current univariate Normal-Normal model to a multivariate version that shrinks buy, sell, and timing dimensions jointly with a covariance structure estimated from data. This replaces the current equal-weighted z-score combination.
- NS2: PnL-weighted posterior:** Replace binary win rate with $\text{PnL} / \sqrt{\text{leverage}}$ as the observable in the hierarchical model. This directly addresses the finding that ROI ($r_s = +0.142$) beats win-rate-based scores ($r_s = +0.109$) because it captures magnitude.
- NS3: Composite score from multiple posteriors (revised priority: now most urgent):** Learn the optimal combination weights for `posterior_buy/sell/timing/sizing/liquidation` via a simple linear model on the train window, rather than equal-weighted z-scores or the literal multiplicative Van Loon formula. **Done (partially) and re-evaluated:** NS4/NS5 below were implemented, but the equal-weighted/multiplicative combination made results *worse* (r_s dropped from $+0.109$ to $+0.058/+0.068$), because `posterior_sizing` is out-of-sample negative. This makes a learned (not equal) combination the single highest-priority next step.
- done* **Add sizing skill as a fourth dimension:** implemented via `fit_continuous_dimension` on $\log(W/L)$ with delta-method sampling variance (§“Bayesian Skill Model”). Result: `posterior_sizing` shrinks moderately (0.325) in-sample but is $r_s = -0.058$ out-of-sample—the dimension does not transfer over a 45-day window and should be down-weighted, not equal-weighted, in any composite.
- done* **Add liquidation rate as a risk-penalty dimension:** implemented as its own rate-style

posterior ($\text{survival_rate} = 1 - \text{liquidation_rate}$). Result: $\text{posterior_liquidation}$ is $r_s = +0.228$ out-of-sample, the strongest score in the entire study, exceeding raw ROI (+0.142). This dimension should be weighted heavily, not just added as a small penalty term as originally scoped.

18.2 Validation Improvements

done **Rolling walk-forward:** implemented as Stage 15 (§16), five non-overlapping 45-day folds with expanding train and per-fold posterior refits. Confirmed all Bayesian scores beat b3_roi by a margin far exceeding the fold std. Extending to more folds with a purged/embargoed split remains open.

NS6: Re-run full comparison (Stage 13): After each model improvement in NS1–NS5, re-run the full baseline comparison; Stage 15 confirms the proposed scores exceed b3_roi ($r_s = 0.173 \pm 0.006$) by $2\text{--}3\times$, well beyond the fold standard deviation.

NS7: Cross-domain validation: Validate the sign-randomisation classifier (B7) on the public Polymarket-v1 dataset (Time Seventeen, 2026, [arXiv:2606.04217](#)) as an independent sanity check before relying on the perp-results for the thesis.

18.3 Longer Horizon (Month 5+)

NS9: Sequence-aware features: Add win-rate trends, streak lengths, and time-varying volatility to capture behavioural change over time. Current features collapse the entire history into summary statistics.

NS10: Regime conditioning: Extend the model with regime-conditional parameters via an HMM on market-level indicators (aggregate PnL, volume volatility). A wallet’s performance in bull regimes may not transfer to bear regimes.

NS11: Crowd-adjustment (RQ3): If follower graph data becomes available (Nansen/GMGN API), estimate crowd-decay function and produce copy-adjusted expected values (Liu et al., 2023; Gao et al., 2026).

NS12: Thesis writing (Month 11–12): Target venue: KDD 2027 or ICAIF 2027.

19 Figures

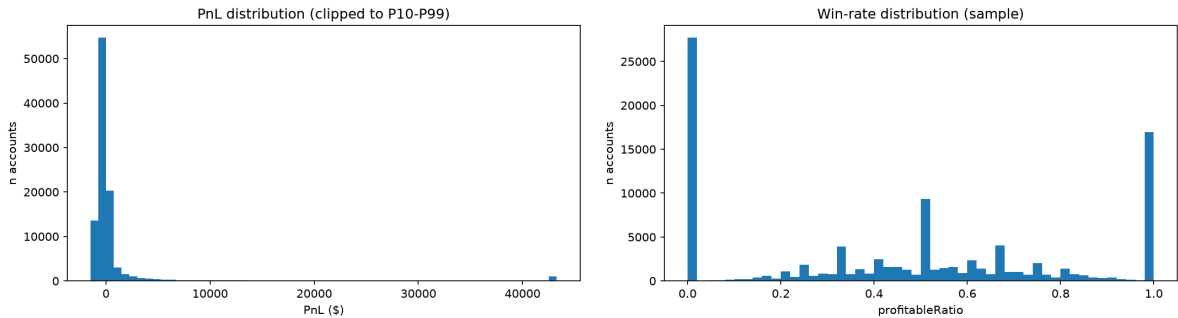


Figure 1: Account-level PnL (left) and win-rate (right) distributions.

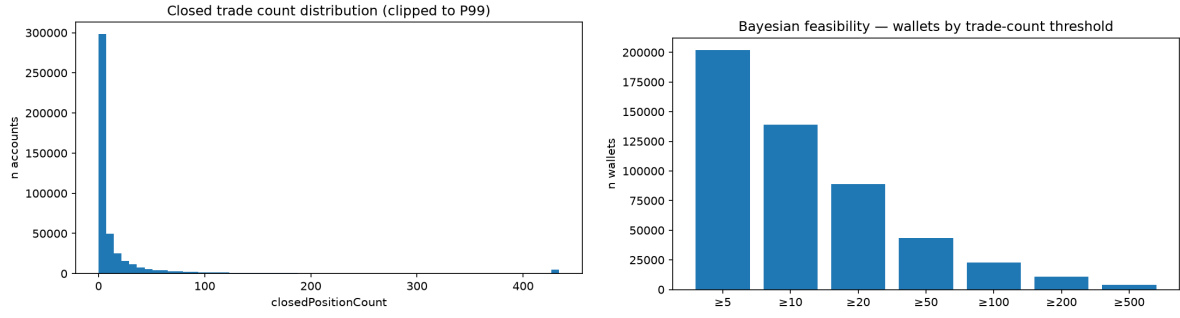


Figure 2: Trade count distribution (left) and Bayesian threshold analysis (right).

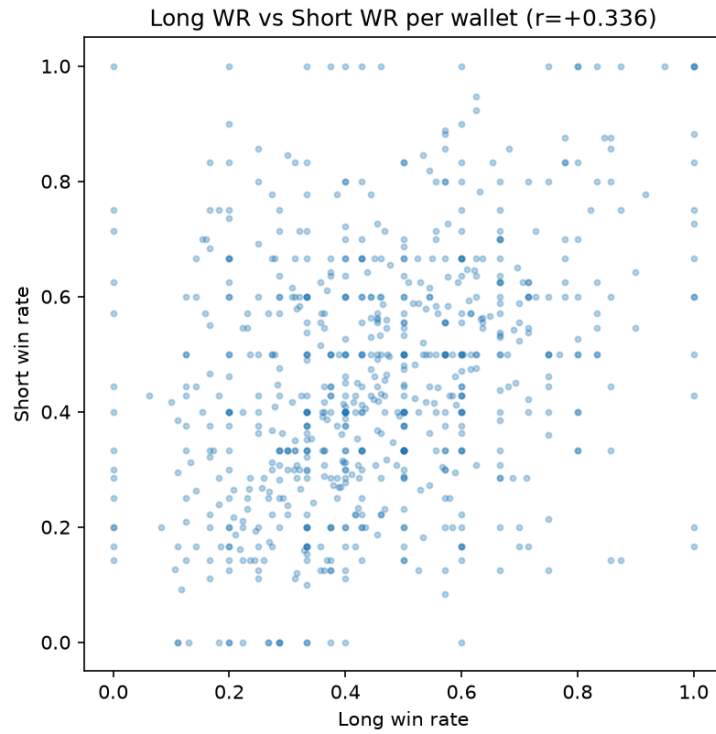


Figure 3: Per-wallet Long win rate vs. Short win rate (n=748 wallets with ≥ 5 trades each side). The Pearson correlation of +0.336 supports side-aware decomposition.

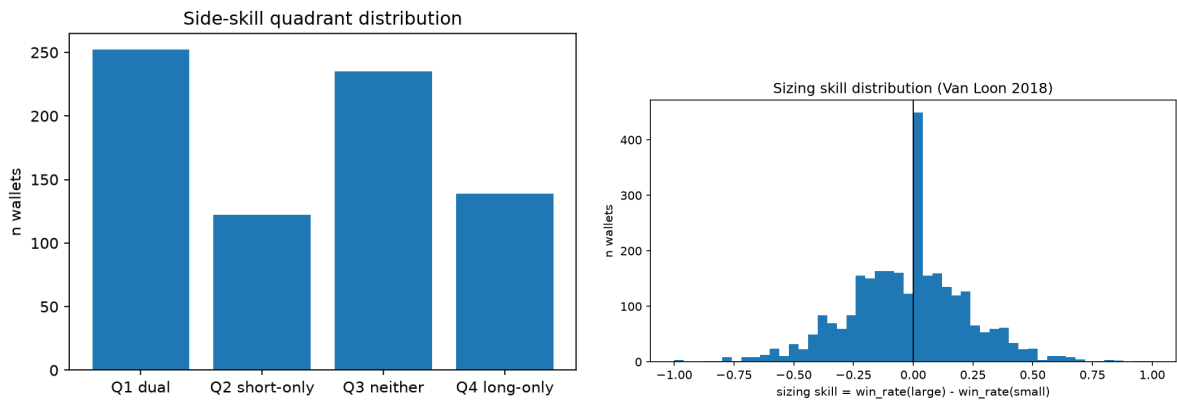


Figure 4: Skill quadrant analysis (left) and sizing skill distribution (right).

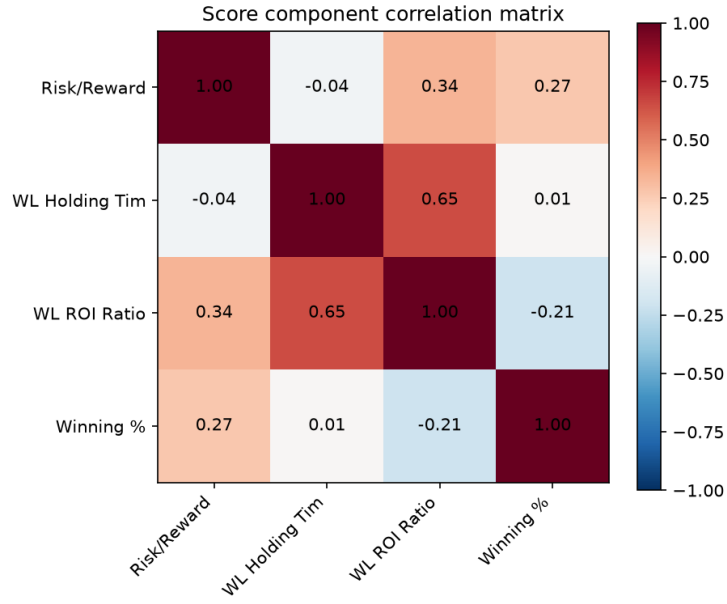


Figure 5: Baseline score component correlation matrix. The strong correlation between WL Holding Time and WL ROI Ratio ($r = 0.652$) indicates redundancy in the current scoring system.

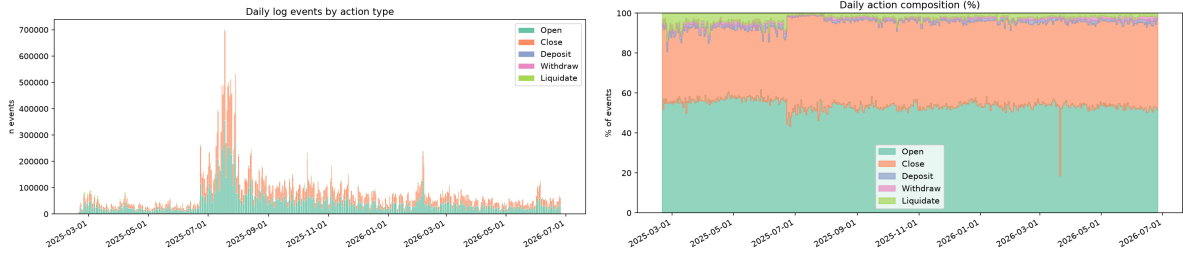


Figure 6: Time series of daily log events (left) and action composition (right).

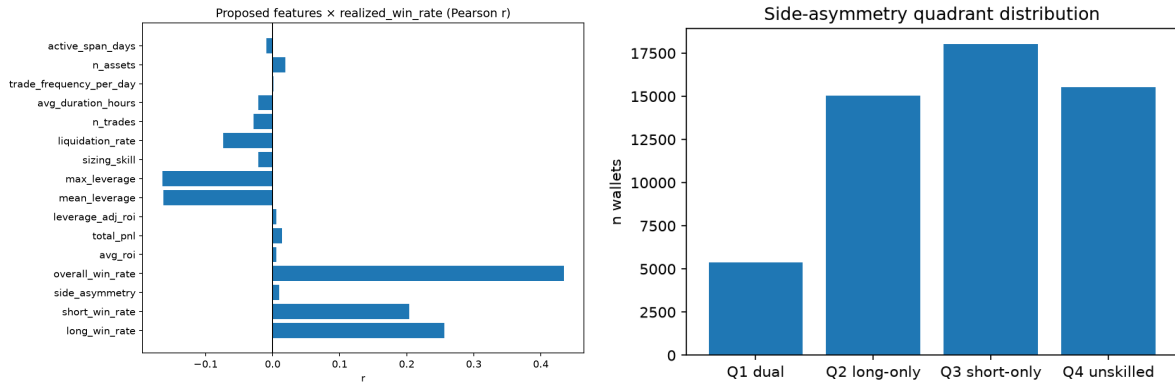


Figure 7: Pipeline feature correlations with realized win rate (left) and quadrant distribution (right).

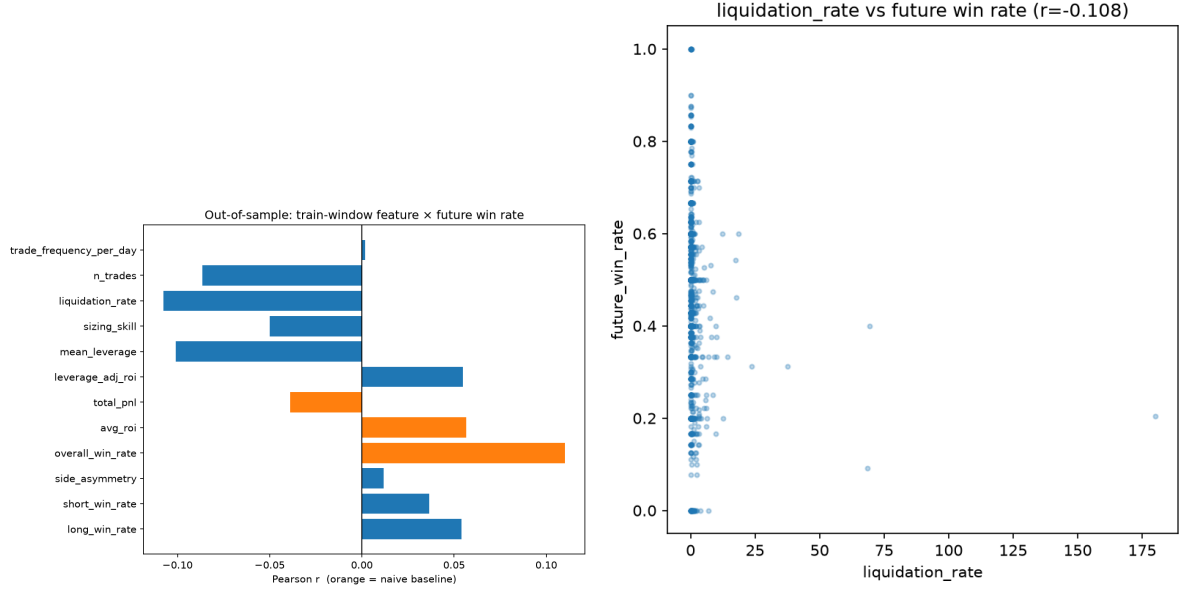


Figure 8: Walk-forward out-of-sample feature correlations (left) and scatter of best proposed feature vs. future win rate (right). Orange bars indicate naive baselines.

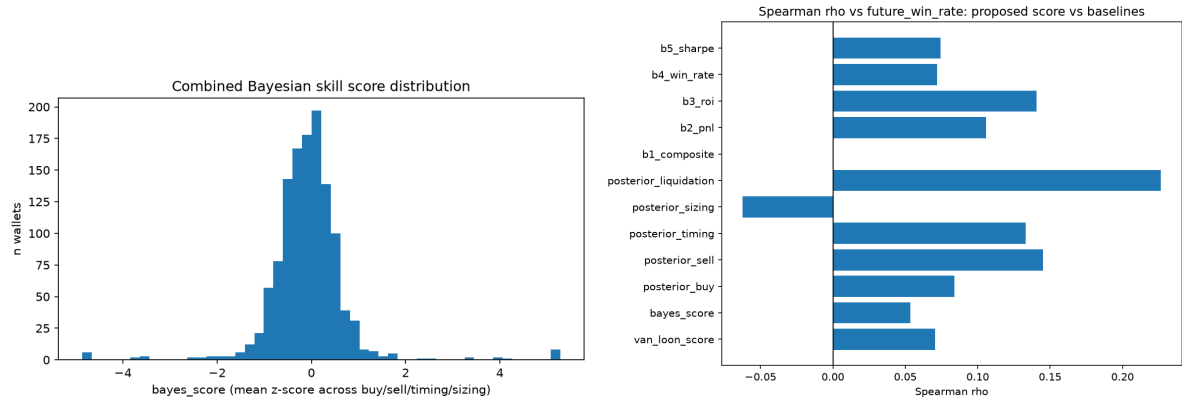


Figure 9: Bayesian score distribution (left) and baseline Spearman rank correlation comparison (right).

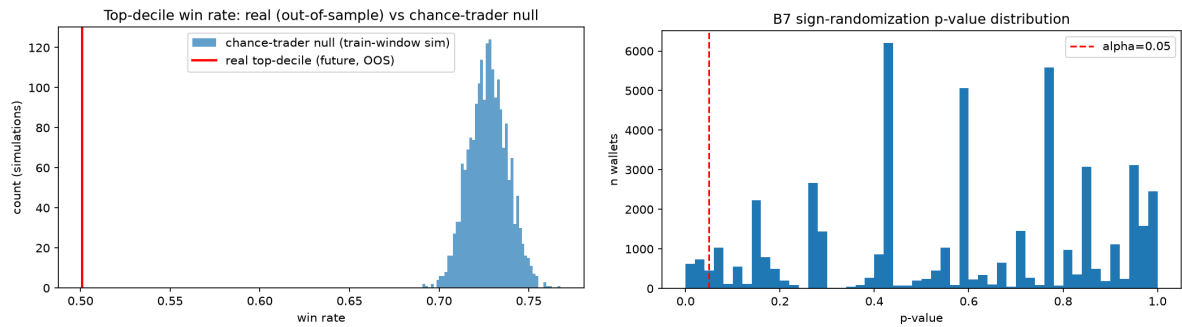


Figure 10: Luck-vs-skill null distribution (left, Papaioannou et al. 2024) and sign-randomisation p -value histogram (right, Gomez-Cram et al. 2026).

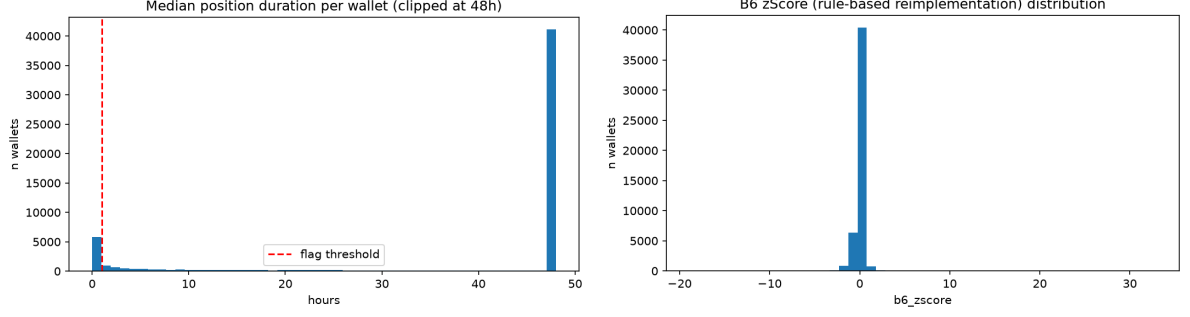


Figure 11: Wash-trade duration distribution (left, Ashfaq 2023) and zScore baseline distribution (right, Anon et al. 2025).

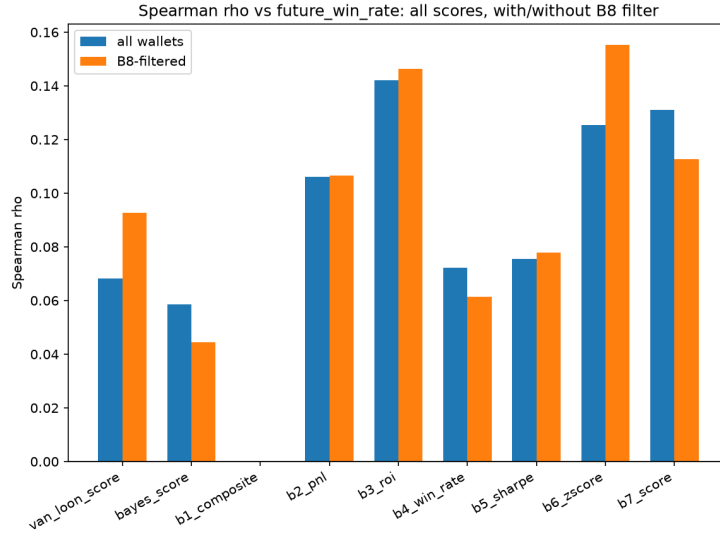


Figure 12: Full baseline comparison: Spearman rank correlation of all eight baselines plus the proposed Bayesian score against out-of-sample future win rate.

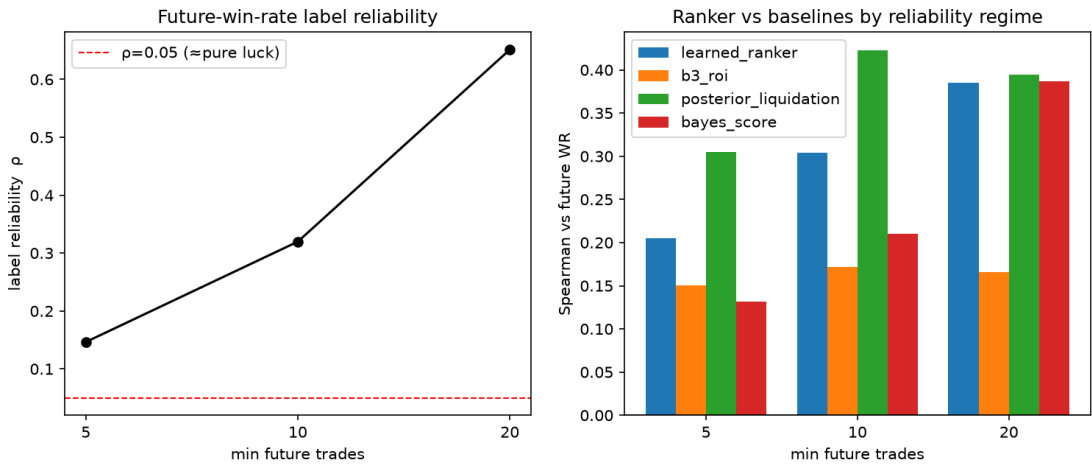


Figure 13: Stage 14. Left: `future_win_rate` label reliability ρ rises sharply with the minimum future-trade threshold; the ≥ 5 evaluation set used in Stages 8–13 sits near the “ $\approx$ pure luck” line. Right: the learned LambdaMART ranker vs. baselines by reliability regime—signal (Spearman vs. future WR) grows monotonically as the label becomes reliable, and the learned ranker beats raw ROI and the equal-weighted `bayes_score` at every level.

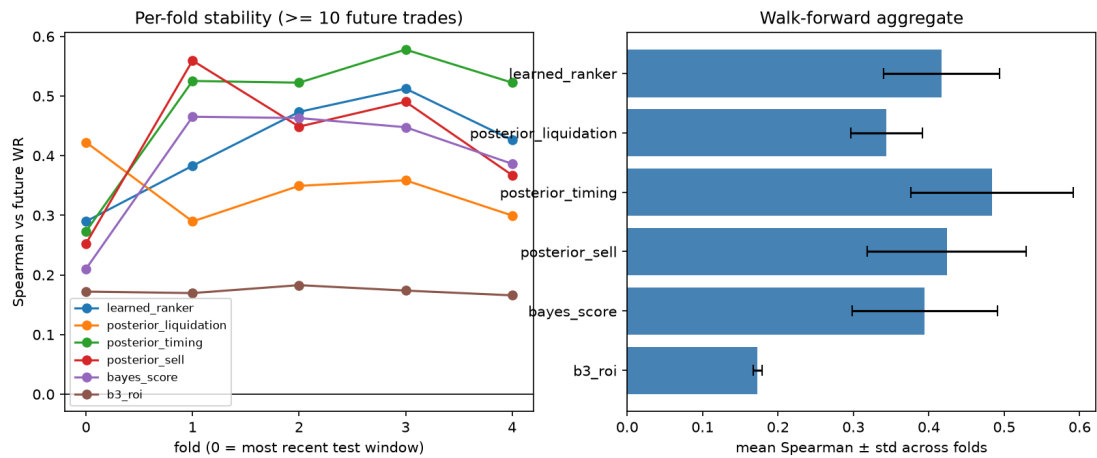


Figure 14: Stage 15 rolling walk-forward (5 folds, expanding train, non-overlapping 45-day test windows). Left: per-fold Spearman vs. future WR—all Bayesian scores are well above raw ROI (bottom flat line) in every fold. Right: aggregate mean $\pm$ std; **posterior_timing** is highest on mean, **learned_ranker** has the smallest error bar among the strong scores, and **b3_roi** is stable but weakest.